



This is low level FPGA interface only - DO NOT USE DIRECTLY - Use RT module to configure

ENCODER SIGNAL

EPTControl

NumberOfCrankTeeth: 7200

Stroke (FALSE=2, TRUE=4): 4

CrankStallPeriod: 5000000

CrankCountStart: 7200

PeriodAccumCount: 7200

SyncEnable: ☐

SimEnable: ☐

MissedCrankFlagClr: ☐

MissedCamFlagClr: ☐

WatchdogIn: ☐

Period: 588

PeriodAccum: 5882000

CrankCount: 5844

CurrentPosition: 23376

MaxCAT: 57600

CrnkSigOS: ☐

CrnkStalled: ☐

SyncStopped: ☐

MissedCrankFlag: ☐

MissedCamFlag: ☐

EPTOutSig

CrnkSigOut: 2

CamSigOut: 0

EncA: 8

EncB: 2

EncC: 2

CamZ: 2

CamZ_filt: ☐

Must Use Cam & Z: ☐

1/2 Z pulse: ☐

toggle TDC: ☐

General Monitoring & Control

Ni9401 Ready: ☐

FPGA Heartbeat: ☐

Chassis Temp Raw (x4 C): 140

dt inj (ticks): 1

dt HB (ms): 500

USER LED is also broadcasting the heartbeat at 1Hz

Additional DO

Mod3/DIO4: ☐

Mod3/DIO5: ☐

Mod3/DIO6: ☐

Mod3/DIO7: ☐

Mod4/DIO4: ☐

Mod4/DIO5: ☐

Mod4/DIO6: ☐

Mod4/DIO7: ☐

DI Driver Settings

DI_Control

ModuleEnable: ☐

WatchdogIn: ☐

OneShotSelect: ☐

OneShotTime: 40000

OneShot: 0

Control: 0

Control2: 128

HostMode: 0

HostCommand: 0

HVTarget: 124

BackBoostTime: 10

Fault_ClsG: ☐

DI_Data_Mod5

ModulePresent: ☐

LVSense: 0

HVSense: 0

HVSense2: 0

TempSense: 0

Fault1: 0

Fault2: 0

OLIntegrator: 0

HardwareRev: 0

FirmwareRev: 0

DI_PhaseInterface

IPhaseDrive (TRUE=HV, FALSE=LV): ☐

IPhaseCurrentUpper: 0

IPhaseCurrentLower: 0

IPhaseDuration: 0

DI Control

DI 1	DI 2	DI 3	DI 4	DI 5	DI 6
Channel Enable: ☐	Channel Enable: ☐	Channel Enable: ☐	Channel Enable: ☐	Channel Enable: ☐	Channel Enable: ☐
Window_Start: 0	Window_Start: 0	Window_Start: 0	Window_Start: 0	Window_Start: 0	Window_Start: 0
Window_End: 0	Window_End: 0	Window_End: 0	Window_End: 0	Window_End: 0	Window_End: 0
Enable_Inj1: ☐	Enable_Inj1: ☐	Enable_Inj1: ☐	Enable_Inj1: ☐	Enable_Inj1: ☐	Enable_Inj1: ☐
Duration_Inj1: 0	Duration_Inj1: 0	Duration_Inj1: 0	Duration_Inj1: 0	Duration_Inj1: 0	Duration_Inj1: 0
Timing_Inj1: 0	Timing_Inj1: 0	Timing_Inj1: 0	Timing_Inj1: 0	Timing_Inj1: 0	Timing_Inj1: 0
Enable_Inj2: ☐	Enable_Inj2: ☐	Enable_Inj2: ☐	Enable_Inj2: ☐	Enable_Inj2: ☐	Enable_Inj2: ☐
Duration_Inj2: 0	Duration_Inj2: 0	Duration_Inj2: 0	Duration_Inj2: 0	Duration_Inj2: 0	Duration_Inj2: 0
Timing_Inj2: 0	Timing_Inj2: 0	Timing_Inj2: 0	Timing_Inj2: 0	Timing_Inj2: 0	Timing_Inj2: 0
Enable_Inj3: ☐	Enable_Inj3: ☐	Enable_Inj3: ☐	Enable_Inj3: ☐	Enable_Inj3: ☐	Enable_Inj3: ☐
Duration_Inj3: 0	Duration_Inj3: 0	Duration_Inj3: 0	Duration_Inj3: 0	Duration_Inj3: 0	Duration_Inj3: 0
Timing_Inj3: 0	Timing_Inj3: 0	Timing_Inj3: 0	Timing_Inj3: 0	Timing_Inj3: 0	Timing_Inj3: 0
Enable_Inj4: ☐	Enable_Inj4: ☐	Enable_Inj4: ☐	Enable_Inj4: ☐	Enable_Inj4: ☐	Enable_Inj4: ☐
Duration_Inj4: 0	Duration_Inj4: 0	Duration_Inj4: 0	Duration_Inj4: 0	Duration_Inj4: 0	Duration_Inj4: 0
Timing_Inj4: 0	Timing_Inj4: 0	Timing_Inj4: 0	Timing_Inj4: 0	Timing_Inj4: 0	Timing_Inj4: 0
Enable_Inj5: ☐	Enable_Inj5: ☐	Enable_Inj5: ☐	Enable_Inj5: ☐	Enable_Inj5: ☐	Enable_Inj5: ☐
Duration_Inj5: 0	Duration_Inj5: 0	Duration_Inj5: 0	Duration_Inj5: 0	Duration_Inj5: 0	Duration_Inj5: 0
Timing_Inj5: 0	Timing_Inj5: 0	Timing_Inj5: 0	Timing_Inj5: 0	Timing_Inj5: 0	Timing_Inj5: 0

Number of Active GN_DI: 0

broadcast inj+ign?: ☐

manual?: ☐

Time out limit: 600000

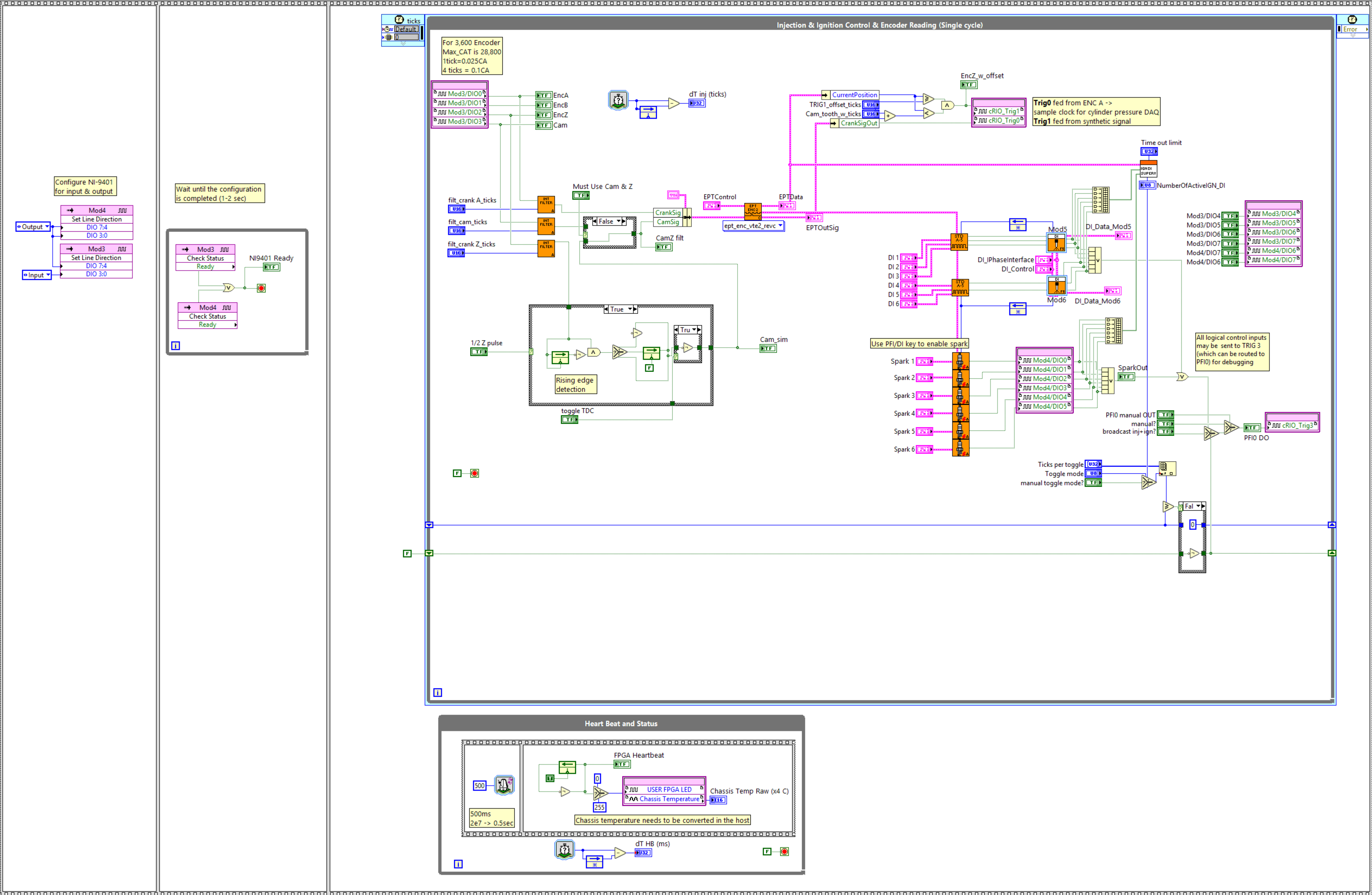
Toggle mode: ☐

manual toggle mode?: ☐

PFI0 manual OUT: ☐

PFI0 DO: ☐

Ticks per toggle: 13000 12000 11000 10000 9000 8000 7000 6000 5000 4000 3000 2000 1000



di_supv_rev5.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Powertrain Controls\Device Drivers\Direct Injector Driver\RevF5\FPGA\Support\di_supv_rev5.vi

filt_int_reva.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Powertrain Controls\Device Drivers\Tools\FPGA\filt_int_reva.vi

ept_revvc.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Powertrain Controls\Device Drivers\Engine Position Tracking\RevC\FPGA\ept_revvc.vi

ept_enc_vtc2_revvc.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Powertrain Controls\Device Drivers\Engine Position Tracking\RevC\Encoder\FPGA\ept_enc_vtc2_revvc.vi

APC_9049_FPGA_IGNDI_supervisor.vi

C:\LabVIEW PROJECT\MONARCH\APC_9049_FPGA_IGNDI_supervisor.vi

PlsGen_Stda5_rev5.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Powertrain Controls\Device Drivers\Direct Injector Driver\RevF5\FPGA\PlsGen_Stda5_rev5.vi

esttl_vt_spark_reva.vi

C:\Program Files (x86)\National Instruments\LabVIEW 2020\vi.lib\Powertrain Controls\Device Drivers\ESTTL\RevA\FPGA\esttl_vt_spark_reva.vi

Position in Hierarchy

